PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

# PANEL 4: The Multiplication Mountain

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'Why does adding physical millimeters multiply the numbers together?'
- Ask: 'Could you build 14 using only two blocks? Which blocks must you choose?'
- Ask: 'What happens to the tower if you try to make 17 out of smaller blocks?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to add the numerals (2 + 3 = 5) instead of reading the product. Direct their attention to the level comparison with Block [6], not Block [5].

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a mystery tower made of three prime blocks. My summit levels exactly with Composite Block [42]. What three pri"
"me blocks am I?"

**OFFICIAL SOLUTION KEY:**

Blocks [2], [3], and [7] (since 2 × 3 × 7 = 42).

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

### INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

### INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

### CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

# PANEL 1: The Multiplication Mountain

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

Can you build multiplication with physical height? In this lab, you discover why multiplying numbers is identical to adding physical tower heights.

**STEP-BY-STEP APPARATUS SETUP:**

1. Pull Prime Blocks [2] (Teal) and [3] (Coral) from your Gridfinity storage tray.
2. Snap Block [3] on top of Block [2] to create a two-block tower.
3. Pull Composite Block [6] (Amber) and place it side-by-side with your tower on the desk.

### REQUIRED HARDWARE & CALIBRATION DATUM

**PARTS NEEDED:** Prime Blocks [2], [3], [5] and Composite Blocks [6], [10], [15] + Stage

**DATUM TARGET:** H(2)=7.00mm + H(3)=11.09mm = H(6)=18.09mm (K_log = 23.2535 mm/dec)

**CALCULATOR KEY:** [2] [ENTER] [3] [×] -> 6

### COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

### 1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

**SINGAPORE DIMENSIONS MATH:** Singapore Dimensions: Grade 4A, Chapter 3 (Factors & Multiples)

**BRIDGES IN MATHEMATICS:** Bridges in Mathematics: Grade 4, Unit 2 (Factors & Prime Numbers)

**BIG IDEAS / CCSS / TEKS:** Big Ideas Math: Grade 6, Sec. 1.2 • CCSS.MATH.4.OA.B.4 • TEKS 5.4A

**GRADE LEVEL BAND:** Grades 4–6 • Target Time: 15 Min

### 2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

**RECOMMENDED COMPANION:** Vol. 1: SC-6 Elementary Companion, Ch. 7 ('Primes & Factor Towers'), pp. 142–156

**HARDWARE UNBOXING BOOKLET:** PRIME-01 Booklet: Page 2 (Log Invariant) & Page 5 (Tutorial 1)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

### CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

# PANEL 2: LAB INVESTIGATION: THE LEVEL SUMMIT TEST

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Lay a straightedge across the summits of Tower [2]+[3] and Block [6]. Notice they level perfectly.

Step 2: Pull Prime Block [5] and snap it on top of Block [2]. Find Composite Block [10] to verify the match.

Step 3: Measure Block [3] and Block [5] stacked. Verify it levels horizontally with Block [15].

Step 4: Record your discovery: Every time you stack prime blocks, their heights ADD together to form the product.

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
H(2)=7.00mm + H(3)=11.09mm = H(6)=18.09mm (K_log = 23.2535 mm/dec)

**PHYSICAL APPARATUS MANIFEST:**

Prime Blocks [2], [3], [5] and Composite Blocks [6], [10], [15] + Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

# PANEL 3: LEVEL 3 CHALLENGE: THE THREE-BLOCK PEAK

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Can you build the number 30 using only prime blocks? How many different orders can you stack them in, and does the final height ever change?

**FORM A MASTER STATION TASKS (Complete All 3):**

**1** **Stack [2] + [3] + [5]. Measure total height against Block [30].**
Expected Physical Finding: Sum = 34.41 mm (Exact Match)

**2** **Stack [5] + [2] + [3] (different order). Does the summit change?**
Expected Physical Finding: Identical height: 34.41 mm

**3** **Execute on SC-6: [2] [ENTER] [3] [×] [5] [×]. What is Register X?**
Expected Physical Finding: 30

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [2] [ENTER] [3] [×] -> 6

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 2 | X: 2 | Key first prime factor into register X |
| ENTER | Y: 2 \| X: 2 | Stack lift copies 2 into register Y |
| 3 | Y: 2 \| X: 3 | Key second prime factor into register X |
| × | X: 6 | Multiply: registers collapse into product 6 |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

# PANEL 4: The Prime Fingerprint

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'Is it possible for any composite number to have two different sets of prime blocks?'
- Ask: 'Why are prime numbers called the bedrock of mathematics?'
- Ask: 'Can you build a tower for 25 using only [2] and [3] blocks? Why not?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students treating 9 as a prime block. Remind them to check if Block [9] can be replaced by two [3] blocks.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A mystery number has the prime fingerprint [2] × [2] × [2] × [3] × [5]. What is the number, and what is its total t"
"ower height?"

**OFFICIAL SOLUTION KEY:**

Number = 120. Tower Height = 3(7.00) + 11.09 + 16.25 = 48.34 mm.

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

### INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

### INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

# PANEL 1: The Prime Fingerprint

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

Every integer greater than 1 has a unique secret code made exclusively of prime numbers. No matter what path you take, you always end up with the exact same pri

**STEP-BY-STEP APPARATUS SETUP:**

1. Pull Composite Block [36] from your kit and place it in the center of the table.
2. Team Member A builds 36 using factor pair 4 × 9 (Stack Block [4] and Block [9]).
3. Team Member B builds 36 using factor pair 6 × 6 (Stack Block [6] and Block [6]).

**REQUIRED HARDWARE & CALIBRATION DATUM**

| | |
|---|---|
| **PARTS NEEDED:** | Prime Blocks [2], [2], [3], [3], [5] and Composite Blocks [12], [18], [36] |
| **DATUM TARGET:** | H(36) = 2·H(2) + 2·H(3) = 14.00 + 22.18 = 36.18 mm |
| **CALCULATOR KEY:** | [2] [x²] [3] [x²] [×] -> 36 |

**COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):**

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

**1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")**

| | |
|---|---|
| **SINGAPORE DIMENSIONS MATH:** | Singapore Dimensions: Grade 5A, Chapter 1, Lesson 1.2 (Prime Factorization) |
| **BRIDGES IN MATHEMATICS:** | Bridges in Mathematics: Grade 5, Unit 1, Module 3 (Factor Trees) |
| **BIG IDEAS / CCSS / TEKS:** | Big Ideas Math: Grade 6, Sec. 1.3 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A |
| **GRADE LEVEL BAND:** | Grades 5–7 • Target Time: 20 Min |

**2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)**

| | |
|---|---|
| **RECOMMENDED COMPANION:** | Vol. 1: SC-6 Elementary Companion, Ch. 7 ('The Prime Fingerprint'), pp. 146–152 |
| **HARDWARE UNBOXING BOOKLET:** | PRIME-01 Booklet: Page 2 (Uniqueness Theorem) & Page 6 (Aha! Moment) |

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

**CLASSROOM LAB STATION RULES:**

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

# PANEL 2: LAB INVESTIGATION: DECOMPOSING TO BEDROCK

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Compare Tower 4 × 9 and Tower 6 × 6. Confirm both level perfectly to Block [36].

Step 2: Decompose Block [4] into two [2] blocks, and Block [9] into two [3] blocks.

Step 3: Decompose both [6] blocks from Team B's tower into [2] and [3].

Step 4: Count bedrock blocks in both piles: Both hold exactly two [2]s and two [3]s ($2^2 \times 3^2$).

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
$H(36) = 2 \cdot H(2) + 2 \cdot H(3) = 14.00 + 22.18 = 36.18$ mm

**PHYSICAL APPARATUS MANIFEST:**

Prime Blocks [2], [2], [3], [3], [5] and Composite Blocks [12], [18], [36]

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

# PANEL 3: LEVEL 3 CHALLENGE: FACTOR TREE DECONSTRUCTION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Can you build Composite Block [60] using three different factor pairs, and prove all three reduce to the exact same prime blocks?

**FORM A MASTER STATION TASKS (Complete All 3):**

1. **Path 1: Start with 6 × 10. Decompose into primes.**
   Expected Physical Finding: [2] + [3] + [2] + [5]

2. **Path 2: Start with 4 × 15. Decompose into primes.**
   Expected Physical Finding: [2] + [2] + [3] + [5]

3. **Path 3: Start with 2 × 30. Decompose into primes.**
   Expected Physical Finding: [2] + [2] + [3] + [5]

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

**CANONICAL KEYSTROKES:** [2] [$x^2$] [3] [$x^2$] [×] -> 36

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 2 | X: 2 | Key base prime 2 |
| $x^2$ | X: 4 | Square prime: $2^2 = 4$ (Tower of two [2] blocks) |
| 3 | Y: 4 \| X: 3 | Key base prime 3 |
| $x^2$ | Y: 4 \| X: 9 | Square prime: $3^2 = 9$ (Tower of two [3] blocks) |
| × | X: 36 | Multiply: 4 × 9 = 36 |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



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PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

# PANEL 4: The Greatest Common Factor

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'If two numbers share no prime blocks in common, what is their GCF?'
- Ask: 'Why can't the GCF ever be taller than the shortest original tower?'
- Ask: 'How does the comparator eliminate guess-and-check errors?'

**⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students including all blocks instead of only shared pairs. Emphasize that GCF requires an exact 1-to-1 matchi
ng exchange between both towers.

**★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Two numbers have a GCF of 14. One number is 42. What is the smallest possible value for the other number?"
""

**OFFICIAL SOLUTION KEY:**

Smallest other number is 14 (or 28, 56, etc.).

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

**INSTANT CLASSROOM ACCESS**

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

**INSTITUTIONAL CURRICULUM CERTIFICATION**

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK)

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

# PANEL 1: The Greatest Common Factor

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

Finding the Greatest Common Factor on paper takes long lists of factors. On the bilateral comparator stage, you simply slide the towers together and pick out the

**STEP-BY-STEP APPARATUS SETUP:**

1. Dock the bilateral slide rule stage onto the center of your table.
2. Build Tower A for 24 using prime blocks: [2], [2], [2], [3].
3. Build Tower B for 36 using prime blocks: [2], [2], [3], [3].

**REQUIRED HARDWARE & CALIBRATION DATUM**

| | |
|---|---|
| **PARTS NEEDED:** | Prime Blocks [2], [2], [2], [3], [3] + Bilateral Slide Rule Comparator Stage |
| **DATUM TARGET:** | GCF(24, 36) = $2^2 \times 3 = 12$ -> Height = 2(7.00) + 11.09 = 25.09 mm |
| **CALCULATOR KEY:** | [24] [ENTER] [36] [GCF] -> 12 |

**COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):**

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

**1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")**

| | |
|---|---|
| **SINGAPORE DIMENSIONS MATH:** | Singapore Dimensions: Grade 5A, Chapter 1, Lesson 1.3 (Greatest Common Factor) |
| **BRIDGES IN MATHEMATICS:** | Bridges in Mathematics: Grade 5, Unit 1, Module 3, Session 2 (GCF) |
| **BIG IDEAS / CCSS / TEKS:** | Big Ideas Math: Grade 6, Sec. 1.4 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A |
| **GRADE LEVEL BAND:** | Grades 5–7 • Target Time: 20 Min |

**2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)**

| | |
|---|---|
| **RECOMMENDED COMPANION:** | Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 152–158 |
| **HARDWARE UNBOXING BOOKLET:** | PRIME-01 Booklet: Page 5 (Comparator Setup) & Page 6 (GCF Proof) |

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

**CLASSROOM LAB STATION RULES:**

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

# PANEL 2: LAB INVESTIGATION: THE OVERLAP EXTRACTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Place Tower A in the left comparator well and Tower B in the right well.

Step 2: Match common blocks pair-by-pair: Pull one [2] from both. Pull a second [2] from both. Pull one [3] from both.

Step 3: What remains? Tower A has an extra [2]; Tower B has an extra [3]. These do not match.

Step 4: Stack the shared matching blocks ([2] + [2] + [3]). Measure this tower against Block [12]. You found the GCF!

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
**GCF(24, 36) = $2^2 \times 3 = 12$ –> Height = 2(7.00) + 11.09 = 25.09 mm**

**PHYSICAL APPARATUS MANIFEST:**

Prime Blocks [2], [2], [2], [3], [3] + Bilateral Slide Rule
Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 3/6 | INTERMEDIATE | 20 Min

# PANEL 3: LEVEL 3 CHALLENGE: THREE-NUMBER COMPARISON

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Can you use prime blocks to find the GCF of 30, 45, and 75 without writing a single calculation on paper?

**FORM A MASTER STATION TASKS (Complete All 3):**

1. **Prime factorize 30, 45, and 75 into block piles.**
   Expected Physical Finding: 30=[2,3,5], 45=[3,3,5], 75=[3,5,5]

2. **Identify the blocks common to ALL THREE piles.**
   Expected Physical Finding: One [3] and one [5]

3. **What is GCF(30, 45, 75)? Verify with Block [15].**
   Expected Physical Finding: 15 (Height = 27.34 mm)

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

**CANONICAL KEYSTROKES:** [24] [ENTER] [36] [GCF] –> 12

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 24 | X: 24 | Key first number into register X |
| ENTER | Y: 24 \| X: 24 | Stack lift |
| 36 | Y: 24 \| X: 36 | Key second number into register X |
| GCF | X: 12 | Extract greatest common divisor from stack |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

# PANEL 4: The Least Common Multiple

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'Why is the LCM always as large as or larger than both starting numbers?'
- Ask: 'If two runners lap a track every 6 and 8 minutes, when do they cross the start line together?'
- Ask: 'How does taking the prime union guarantee the smallest leveling summit?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking LCM means multiplying 6 × 8 = 48. Show them that 24 levels first, making 24 the LEAST common multiple.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Traffic light A flashes every 12 seconds. Traffic light B flashes every 18 seconds. Traffic light C flashes every 3"
"0 seconds. When do all three flash together?"

**OFFICIAL SOLUTION KEY:**

LCM(12, 18, 30) = $2^2 \times 3^2 \times 5 = 180$ seconds (3 minutes).

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

**INSTANT CLASSROOM ACCESS**

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

**INSTITUTIONAL CURRICULUM CERTIFICATION**

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

# PANEL 1: The Least Common Multiple

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

When will two repeating rhythmic patterns clash together at the exact same beat? In this lab, you build stepping towers of 6 and 8 until their summits level off.

**STEP-BY-STEP APPARATUS SETUP:**

1. Pull several Block [6]s and Block [8]s from your storage tray.
2. Team Member A stacks blocks of 6: 6, 12, 18, 24, 30...
3. Team Member B stacks blocks of 8: 8, 16, 24, 32...

**REQUIRED HARDWARE & CALIBRATION DATUM**

| | |
|---|---|
| **PARTS NEEDED:** | Prime Blocks [2], [2], [2], [3], [3] + Multiples of 6 and 8 |
| **DATUM TARGET:** | LCM(6, 8) = $2^3 \times 3 = 24$ -> Height = 3(7.00) + 11.09 = 32.09 mm |
| **CALCULATOR KEY:** | [6] [ENTER] [8] [LCM] -> 24 |

**COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):**

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

**1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")**

| | |
|---|---|
| **SINGAPORE DIMENSIONS MATH:** | Singapore Dimensions: Grade 6A, Chapter 1, Lesson 1.4 (Least Common Multiple) |
| **BRIDGES IN MATHEMATICS:** | Bridges in Mathematics: Grade 5, Unit 1, Module 3, Session 3 (LCM) |
| **BIG IDEAS / CCSS / TEKS:** | Big Ideas Math: Grade 6, Sec. 1.5 • CCSS.MATH.6.NS.B.4 • TEKS 6.7A |
| **GRADE LEVEL BAND:** | Grades 5–7 • Target Time: 20 Min |

**2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)**

| | |
|---|---|
| **RECOMMENDED COMPANION:** | Vol. 1: SC-6 Elementary Companion, Ch. 7, pp. 156–162 |
| **HARDWARE UNBOXING BOOKLET:** | PRIME-01 Booklet: Page 2 (LCM Formula) & Page 6 (Aha! Discovery) |

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

**CLASSROOM LAB STATION RULES:**

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

# PANEL 2: LAB INVESTIGATION: THE SUMMIT LEVELING RACE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Build Tower 6 (height 18.09mm) and Tower 8 (height 21.00mm). Notice Tower 8 is taller.

Step 2: Add another [6] to Tower 6. Now Tower 6 is 12 (height 25.09mm) and taller than 8.

Step 3: Add another [8] to Tower 8. Tower 8 is 16 (height 28.00mm). Keep going!

Step 4: Stop when both summits reach the exact same horizontal level: 24 (height 32.09mm). You found LCM(6,8)!

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
**LCM(6, 8) = $2^3 \times 3 = 24$ –> Height = 3(7.00) + 11.09 = 32.09 mm**

**PHYSICAL APPARATUS MANIFEST:**

Prime Blocks [2], [2], [2], [3], [3] + Multiples of 6 and 8

**💡 LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 4/6 | INTERMEDIATE | 20 Min

# PANEL 3: LEVEL 3 CHALLENGE: THE PRIME UNION SHORTCUT

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Instead of stacking composite blocks, can you find LCM(12, 18) in 10 seconds by taking the UNION of their prime blocks?

**FORM A MASTER STATION TASKS (Complete All 3):**

**1** **Factor 12: [2, 2, 3]. Factor 18: [2, 3, 3].**
Expected Physical Finding: **Blocks: two [2]s, two [3]s**

**2** **Combine the maximum count of each prime: [2, 2, 3, 3].**
Expected Physical Finding: **$2^2 \times 3^2 = 36$**

**3** **Verify: Does 36 level both Tower 12 and Tower 18?**
Expected Physical Finding: **Yes! 12×3=36 and 18×2=36.**

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

**CANONICAL KEYSTROKES:** [6] [ENTER] [8] [LCM] –> 24

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 6 | X: 6 | Key first number |
| ENTER | Y: 6 \| X: 6 | Stack lift |
| 8 | Y: 6 \| X: 8 | Key second number |
| LCM | X: 24 | Calculate least common multiple |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



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PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

# PANEL 4: The Product-Balance Theorem

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'Where did the blocks from Tower A and Tower B go when we made GCF and LCM?'
- Ask: 'Can any prime block be created or destroyed during this regrouping?'
- Ask: 'Why does this prove that GCF(A,B) × LCM(A,B) = A × B for ALL numbers?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking this is an approximation. Emphasize that because prime blocks are physical conserved units, this is an exact geometric proof.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Two numbers have a product of 1,200. Their GCF is 10. Without knowing what the numbers are, what is their LCM?"
""

---

**OFFICIAL SOLUTION KEY:**

LCM = 1,200 / 10 = 120.

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

**INSTANT CLASSROOM ACCESS**

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

**INSTITUTIONAL CURRICULUM CERTIFICATION**

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- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

# PANEL 1: The Product-Balance Theorem

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

Is it a coincidence that multiplying the GCF and LCM always equals the product of the original two numbers? In this lab, you prove this universal theorem using p

**STEP-BY-STEP APPARATUS SETUP:**

1. Choose numbers A = 12 and B = 18.
2. Build Tower A (12 = [2]+[2]+[3]) and Tower B (18 = [2]+[3]+[3]).
3. Stack Tower A on top of Tower B to create the 'Combined Product Tower' (12 × 18).

**REQUIRED HARDWARE & CALIBRATION DATUM**

| | |
|---|---|
| **PARTS NEEDED:** | Prime Blocks [2], [2], [2], [3], [3] for numbers 12 and 18 + Comparator Stage |
| **DATUM TARGET:** | H(12) + H(18) = H(GCD) + H(LCM) = H(6) + H(36) = 54.27 mm |
| **CALCULATOR KEY:** | [12] [ENTER] [18] [×] [12] [ENTER] [18] [GCF] [12] [ENTER] [18] [LCM] [×] [−] -> 0 |

**COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):**

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

**1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")**

| | |
|---|---|
| **SINGAPORE DIMENSIONS MATH:** | Singapore Dimensions: Grade 6A, Chapter 1, Lesson 1.5 (Advanced Number Theory) |
| **BRIDGES IN MATHEMATICS:** | Bridges in Mathematics: Grade 5, Unit 1, Module 3 (Summit Balances) |
| **BIG IDEAS / CCSS / TEKS:** | Big Ideas Math: Grade 6, Ch. 1 Review • CCSS.MATH.6.NS.B.4 • TEKS 6.7A |
| **GRADE LEVEL BAND:** | Grades 6–8 • Target Time: 25 Min |

**2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)**

| | |
|---|---|
| **RECOMMENDED COMPANION:** | Vol. 2: SC-32 Scientific Handbook, Ch. 7 ('Brent Solver & Number Theory'), pp. 88–96 |
| **HARDWARE UNBOXING BOOKLET:** | PRIME-01 Booklet: Page 2 (Theorem 1.4) & Page 6 (Aha! Discovery) |

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

**CLASSROOM LAB STATION RULES:**

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

# PANEL 2: LAB INVESTIGATION: THE PRIME CONSERVATION PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Pull out the GCF blocks shared by both numbers: one [2] and one [3] (Tower GCF = 6).

Step 2: Collect all the remaining blocks into the LCM pile: [2], [2], [3], [3] (Tower LCM = 36).

Step 3: Stack Tower GCF (6) on top of Tower LCM (36) to create the 'GCF × LCM Tower'.

Step 4: Stand the Combined Product Tower beside the GCF × LCM Tower: Both towers contain the exact same five prime blocks! They level perfectly!

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
**H(12) + H(18) = H(GCD) + H(LCM) = H(6) + H(36) = 54.27 mm**

**PHYSICAL APPARATUS MANIFEST:**

Prime Blocks [2], [2], [2], [3], [3] for numbers 12 and 18 +
Comparator Stage

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 5/6 | ADVANCED | 25 Min

# PANEL 3: LEVEL 3 CHALLENGE: THE ALGEBRAIC EQUALITY

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Test this invariant with numbers 20 and 30. Does H(20) + H(30) equal H(GCF) + H(LCM)?

**FORM A MASTER STATION TASKS (Complete All 3):**

1. **Find GCF(20, 30) and LCM(20, 30).**
   Expected Physical Finding: GCF = 10, LCM = 60

2. **Calculate 20 × 30 and 10 × 60.**
   Expected Physical Finding: Both equal 600

3. **Count prime blocks: 20=[2,2,5], 30=[2,3,5]. What are GCF and LCM primes?**
   Expected Physical Finding: GCF=[2,5], LCM=[2,2,3,5]. Total primes conserved!

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

**CANONICAL KEYSTROKES:** [12] [ENTER] [18] [×] [12] [ENTER] [18] [GCF] [12] [ENTER] [18] [LCM] [×] [−] -> 0

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 12 | X: 12 | Key A = 12 |
| ENTER | Y: 12 \| X: 12 | Stack lift |
| 18 | Y: 12 \| X: 18 | Key B = 18 |
| × | X: 216 | A × B = 216 |
| 6 | Y: 216 \| X: 6 | Key GCF(12, 18) = 6 |
| 36 | Y: 216 \| X: 6 \| X: 36 | Key LCM(12, 18) = 36 |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



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PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

# PANEL 4: Exponents, Stacks & Cryptography

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

## TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

**SOCRATIC PROMPTS (Ask Your Table Team):**

- Ask: 'Why did finding 13 × 17 take only 1 second, but finding the factors of 221 took multiple minutes?'
- Ask: 'What happens when the prime numbers have 300 digits instead of 2 digits?'
- Ask: 'How does the RPN stack mirror building a tower of blocks?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students testing even numbers like 8 or 10. Remind them: if a number is not divisible by 2, it CANNOT be divis
ible by any multiple of 2!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"I am a semiprime number (product of exactly two primes). My smaller prime factor is 11. My total tower height is 38"
".64 mm. What number am I?"

---

**OFFICIAL SOLUTION KEY:**

H(11) = 24.21 mm. Remaining height = 38.64 − 24.21 = 14.43 mm -> H(13) = 25.90 mm... Testing: 11 × 23 = 253.

## 3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

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step-by-step video solution, and printable rubric.

https://stackcalc.io/products/prime-studio/#product-videos

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- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.



---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

# PANEL 1: Exponents, Stacks & Cryptography

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

## STATION CHALLENGE HOOK & LAB TABLE SETUP

**THE MISSION:**

Modern internet banking and cybersecurity (RSA cryptography) are built on a simple secret: multiplying two prime numbers together is instant, but factoring them

**STEP-BY-STEP APPARATUS SETUP:**

1. Pull Prime Blocks [17] and [19] from your storage tray.
2. Multiply them together on your SC-32 calculator: [17] [ENTER] [19] [×]. (Result: 323).
3. Challenge: If someone gives you the number 323, how hard is it to find the two prime blocks without a computer?

**REQUIRED HARDWARE & CALIBRATION DATUM**

| | |
|---|---|
| **PARTS NEEDED:** | Full Prime Kit (Blocks 2, 3, 5, 7, 11, 13, 17, 19) + SC-32 Hardware |
| **DATUM TARGET:** | Prime Tower H(p^k) = k · H(p) (Direct physical multiplication of height) |
| **CALCULATOR KEY:** | [2] [ENTER] [5] [y^x] -> 32 |

**COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):**

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

## TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

**1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")**

| | |
|---|---|
| **SINGAPORE DIMENSIONS MATH:** | Singapore Dimensions: Grade 7A, Chapter 1 (Prime Numbers & Exponents) |
| **BRIDGES IN MATHEMATICS:** | Bridges in Mathematics: Grade 5, Unit 8 (Culminating Math Projects) |
| **BIG IDEAS / CCSS / TEKS:** | Big Ideas Math: Accelerated Grade 7, Ch. 1 • CCSS.MATH.8.EE.A.1 • TEKS 8.2D |
| **GRADE LEVEL BAND:** | Grades 6–8+ • Target Time: 25 Min |

**2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)**

| | |
|---|---|
| **RECOMMENDED COMPANION:** | Vol. 2: SC-32 Scientific Handbook, Ch. 10 ('Computer Science, Bases & Ciphers'), pp. 160–174 |
| **HARDWARE UNBOXING BOOKLET:** | PRIME-01 Booklet: Page 7 (RPN Stack Connection) & Page 8 (Exercises) |

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

**CLASSROOM LAB STATION RULES:**

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

# PANEL 2: LAB INVESTIGATION: EXPONENTIAL REPETITION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

## LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

**STEP-BY-STEP LABORATORY EXPLORATION:**

Step 1: Build a tower of five [2] blocks ($2^5$). Measure height: 5 × 7.00 = 35.00 mm.

Step 2: Key [2] [ENTER] [5] [y^x] into SC-32. Verify Register X reads 32.

Step 3: Build a tower of three [3] blocks ($3^3 = 27$). Height = 3 × 11.09 = 33.27 mm.

Step 4: Combine $2^3 \times 3^2$ (three [2]s and two [3]s). Verify Register X reads 72.

**KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:**

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

## CALIBRATION & DATUM TARGET

**MITUTOYO ±0.05 MM CALIBRATION**

Target Datum:
**Prime Tower H(p^k) = k · H(p) (Direct physical multiplication of height)**

**PHYSICAL APPARATUS MANIFEST:**

Full Prime Kit (Blocks 2, 3, 5, 7, 11, 13, 17, 19) + SC-32 H
ardware

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk.
The snap detents are engineered for a clean 6N tactile
insertion force. If parts wedge or tilt, inspect
the socket floor for stray dust or misalignment.

**Level check: Eye-level sighting is accurate to ±0.2 mm.**



---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

PRIME-01 • TENT 6/6 | ADVANCED | 25 Min

# PANEL 3: LEVEL 3 CHALLENGE: THE RSA ONE-WAY FUNCTION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

## LEVEL 3 CHALLENGE: THE "AHA!" DISCOVERY MOMENT

**THE NON-ROUTINE PUZZLE:**

Can you find the two prime factor blocks that multiply to make 221? (Hint: Neither factor is 2, 3, or 5).

**FORM A MASTER STATION TASKS (Complete All 3):**

1. **Test prime 7: Does 221 divide by 7?**
   Expected Physical Finding: **221 ÷ 7 = 31.57 (No)**

2. **Test prime 11: Does 221 divide by 11?**
   Expected Physical Finding: **221 ÷ 11 = 20.09 (No)**

3. **Test prime 13: Does 221 divide by 13?**
   Expected Physical Finding: **221 ÷ 13 = 17 (Exact match: 13 × 17 = 221!)**

## RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

**CANONICAL KEYSTROKES:** [2] [ENTER] [5] [y^x] -> 32

| KEY | REGISTER STATE (T, Z, Y, X) | MATHEMATICAL STACK ACTION |
|---|---|---|
| 2 | X: 2 | Base prime 2 |
| ENTER | Y: 2 \| X: 2 | Stack lift |
| 5 | Y: 2 \| X: 5 | Exponent 5 (Stack of five [2] blocks) |
| y^x | X: 32 | Evaluate $2^5 = 32$ |

**CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:**

• The physical manipulative is not separate from the calculator—it is an exact physical mirror.
• Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
• Pressing an operator key ([+], [×], [−], [/]) corresponds to leveling two blocks together.
• In standard testing, students who mastered the physical blocks recall the register stack
spatially, making parentheses syntax errors completely impossible.

**Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.**



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